

Outbound Funnel Intelligence: Uncoordinated Multi-Touch Outreach Analysis

1. Project Overview

Business Context

Outbound sales teams often engage the same prospect through multiple campaigns, senders, and touchpoints. While this increases outreach coverage, it can also create duplicate engagement, ownership conflicts, and unreliable funnel reporting.

What This Project Analyzes

- Prospect journeys from outreach to conversion
- Funnel drop-offs across key sales stages
- Sender overlap and outreach coordination
- CRM duplicate identity impact

Business Value

- Established a journey-level KPI framework
- Identified funnel bottlenecks and conversion leakage
- Quantified sender overlap and CRM duplication impact
- Supported data-driven outreach optimization

2. Business Problem

Key Challenges

- **Multiple senders contacting the same prospect created overlap and inconsistent outreach.**
- **Limited visibility into prior interactions reduced follow-up effectiveness.**
- **CRM duplicate identities affected reporting accuracy and attribution.**
- **Funnel progression and lead status were not consistently measurable.**
- **Sales teams lacked a reliable framework for lead prioritization.**

Business Impact

These issues reduced reporting reliability, increased operational inefficiencies, and limited visibility into true outreach performance.

Primary Objectives

Develop a trusted journey-level framework to:

- Measure outreach effectiveness consistently
- Identify sender overlap and ownership issues
- Improve reporting reliability
- Support better lead prioritization
- Enable accurate funnel performance analysis

3. Project Objectives

Project Goal

- Track prospect journeys from outreach to conversion using a consistent journey-level framework.
- Identify key funnel bottlenecks and conversion leakage points.
- Measure the impact of sender overlap on outreach performance.
- Evaluate the effect of CRM duplicate identities on reporting and coordination.
- Analyze outreach efficiency across different touchpoint levels.

Key Business Questions

- Where does the largest funnel drop-off occur?
- What impact does sender overlap have on performance?
- How do duplicate identities affect reporting accuracy?
- When do additional touchpoints become less effective?
- Which response stages are most likely to convert?

4. Dataset Overview

The project uses five operational datasets representing outreach activity, engagement behavior, campaign information, sender ownership, and contact-level CRM records. Together, these datasets capture the complete prospect journey from initial outreach through qualification and conversion.

Dataset	Grain	Purpose
Email Activity	1 row = 1 outreach event	Stores outbound email activity, sequence progression, and touchpoint history
Engagement	1 row = 1 engagement event	Stores responses, qualification outcomes, and final conversion status

Contacts	1 row = 1 CRM contact record	Stores prospect attributes and identity information
Campaigns	1 row = 1 campaign	Stores campaign configuration and outreach strategy details
Senders	1 row = 1 sender	Stores sender ownership and segmentation information

Data Considerations

During validation, duplicate human identities were identified across multiple CRM contact records. Rather than removing these records, they were preserved for analysis because they represented an operational visibility and attribution challenge that directly influenced outreach coordination and reporting reliability.

5. Data Preparation & Processing

5.1 Excel / Power Query

- Imported source datasets for initial validation and preparation.
- Standardized data types, reviewed missing values, and validated response-related fields.
- Created a deduplicated contact table during exploration to assess CRM identity quality.
- Identified duplicate CRM identities as a meaningful operational characteristic rather than a data quality issue.
- Retained original contact records to preserve analytical accuracy.

5.2 MySQL Data Validation & Modeling

- Loaded datasets into MySQL for schema standardization and data quality validation.
- Introduced duplicate identity indicators to measure CRM duplication without removing records.
- Validated relationships across outreach, engagement, campaign, sender, and contact datasets.
- Built analytical summary layers for:
 - Funnel measurement
 - Sender overlap analysis
 - Outreach efficiency analysis
 - KPI reporting
- Consolidated validated analytical layers into a final analytical dataset for reporting and dashboard development.

5.3 Power BI Modeling

- Connected Power BI to the final analytical dataset and campaign reference data.

- Developed KPI measures using a validated journey-level reporting grain.
- Enabled consistent measurement of:
 - Outreach activity
 - Qualification and conversion performance
 - Sender overlap
 - Outreach efficiency
 - CRM data quality metrics

6. Analytical Modeling Decisions

1. CRM Duplicate Identities Were Preserved

Initial exploration included a deduplicated contact table to assess CRM data quality. Further validation showed that duplicate contact records represented a real operational challenge rather than a data-cleaning issue. To preserve analytical accuracy, duplicate identities were retained and later analyzed using dedicated duplicate indicators.

2. Response Classification Was Standardized

The original response categories contained multiple variations that were not suitable for consistent analysis. A standardized response classification layer was created to support reliable measurement of engagement outcomes while preserving the original source values.

3. Qualification Logic Was Explicitly Defined

A dedicated qualification flag was introduced because qualification outcomes were central to funnel analysis. Based on dataset validation and business definitions, responses indicating direct requirements or meeting interest were classified as qualified opportunities, creating a consistent qualification framework for downstream reporting.

4. Email Was Selected as the Human Identity Anchor

Multiple CRM contact records could represent the same individual. Email addresses provided the most reliable representation of a unique human identity across the dataset and enabled prospect-level analysis without losing visibility into CRM duplication.

5. Email Activity Was Used as the Primary Fact Table

Email Activity was selected as the analytical foundation because it captured the complete outreach history, including campaign participation, sender activity, touchpoints, and sequence progression. Every major business process in the funnel originated from outreach activity, making it the most suitable base table for modeling.

6. Relationship Validation Was Performed Before Modeling

Relationships between outreach, engagement, campaign, sender, and contact datasets were validated before KPI development. This ensured analytical consistency and reduced the risk of inaccurate reporting caused by invalid joins or relationship assumptions.

7. Summary Layers Were Created to Resolve Grain Differences

The Email Activity and Engagement datasets operated at different grains and contained many-to-many relationship characteristics. To support reliable analysis, dedicated summary layers were created for engagement outcomes, outreach touchpoints, and sender overlap calculations before final model construction.

8. Sender Overlap Was Measured at the Human Identity Level

Sender overlap was evaluated using email-based identities rather than CRM contact records. Measuring overlap at the contact level would have understated the true coordination challenge because multiple contact records could belong to the same individual.

9. Analytical Master Was Maintained at the Event Level

The final analytical dataset was intentionally preserved at the outreach-event level to maintain operational detail and analytical flexibility. This allowed business questions to be analyzed without losing visibility into individual outreach activities.

10. KPI Calculations Were Standardized at the Journey Level

While the analytical model remained at the event level, KPI calculations were controlled using a journey grain defined as Email + Campaign. This approach prevented metric inflation caused by multiple outreach events and ensured consistent measurement of reply, qualification, conversion, sender overlap, and outreach efficiency metrics.

7. KPI Framework

KPI Design Principles

All KPI calculations were standardized at the journey level using **Email + Campaign** as the reporting grain. While the analytical model remained at the outreach-event level, KPI calculations were intentionally controlled at the journey level to prevent inflation caused by multiple outreach events, duplicate CRM identities, and many-to-many relationship characteristics within the source data.

This approach ensured that funnel performance, sender overlap, outreach efficiency, and CRM data quality metrics were measured consistently across the project.

KPI Category	KPIs Measured
Funnel Performance	Total Journeys, Replied Journeys, Qualified Journeys, Converted Journeys, Reply Rate, Qualified Rate, Conversion Rate
Funnel Progression	Reply-to-Qualification Rate, Qualification-to-Conversion Rate
Sender Overlap Analysis	Sender Overlap Rate, Overlap Conversion Rate, No Overlap Conversion Rate, Conversion Delta
Outreach Efficiency	Average Touchpoints, Average Touchpoints (Converted), Average Touchpoints (Non-Converted), Average Outreach Duration
CRM Data Quality	CRM Duplication Rate, Duplicate Identity Conversion Rate, Single Identity Conversion Rate, Duplicate Identity Sender Overlap Rate
Response Quality	Direct Requirement Conversion Rate, Meeting Interest Conversion Rate, Negative Response Rate

8. Dashboard Overview

Page 1 — Executive Funnel Summary: Overall funnel performance and stage-wise leakage.

Page 2 — Sender Overlap Impact Analysis: Operational impact of multi-sender outreach.

Page 3 — Outreach Efficiency: Touchpoint effectiveness and diminishing returns.

Page 4 — Response Quality & CRM Data Quality Impact: Response-stage conversion and CRM duplication analysis.

9. Key Findings

Funnel Performance

- The outbound funnel achieved a **30.44% Reply Rate** and **3.99% Conversion Rate** at the journey level.
- The largest funnel leakage occurred between the Reply and Qualification stages, with a **71.2% drop-off rate**, indicating that initial engagement alone was not a strong indicator of sales readiness.
- Qualified journeys converted at **45.5%**, making qualification quality the strongest predictor of downstream conversion performance.

Sender Overlap Analysis

- Sender overlap was present in **72.5% of prospect journeys**, highlighting significant coordination exposure across outreach activities.
- Conversion performance between overlapped and non-overlapped journeys differed by only **0.27 percentage points**, indicating minimal direct impact on conversion outcomes.
- Findings suggest that sender overlap is primarily an operational coordination and ownership challenge rather than a performance problem.

Outreach Efficiency

- Conversion efficiency was strongest at the **first touchpoint (8.03%)** and generally declined as additional outreach attempts were introduced.
- Converted journeys required an average of **2.93 touchpoints**, compared to **3.99 touchpoints** for non-converted journeys.
- Extended outreach sequences generated diminishing returns, with later touchpoints contributing significantly less conversion value than early-stage outreach.

CRM Data Quality & Response Analysis

- CRM duplicate identities accounted for **47.37% of CRM records**, exposing significant reporting and ownership challenges.

- Duplicate identities experienced **96.94% sender overlap**, compared to **34.87% for single identities**, indicating substantially higher outreach exposure.
- Direct Requirement (**51.56%**) and Meeting Interest (**40.54%**) responses demonstrated the highest conversion rates among all response stages, making them the strongest indicators of future conversion.

10. Business Recommendations

1. Implement identity-level CRM deduplication controls to improve ownership visibility, reporting reliability, and outreach coordination.
2. Prioritize high-intent response signals such as Direct Requirement and Meeting Interest during lead qualification and follow-up processes.
3. Establish cross-campaign ownership and outreach governance to reduce unnecessary sender overlap while maintaining prospect coverage.

11. Limitations & Assumptions

1. Analysis was performed using historical outreach and engagement data.
2. Identity resolution was based on email-level matching across datasets.
3. External business factors influencing conversion were not available within the dataset.
4. Results should be interpreted within the scope of the provided CRM and outreach data.

12. Technical Stack

- Excel: Initial data exploration, quality assessment, and validation of source datasets.
- Power Query: Data preparation, transformation, standardization, and loading workflows.
- SQL (MySQL): Data cleaning, relationship validation, analytical modeling, summary layer creation, and KPI development.
- Power BI: Interactive dashboard development, data visualization, and executive reporting.
- DAX: Journey-level KPI calculations, funnel metrics, and reporting measures.
- GitHub: Project documentation, repository management, and version control.

13. Conclusion

This project established a validated journey-level analytical framework for measuring outbound funnel performance, outreach efficiency, sender overlap, and CRM data quality. The analysis demonstrated that operational visibility and identity management challenges had a greater impact on reporting reliability than on conversion performance, providing actionable opportunities for process improvement and outreach optimization.